



University of Asia Pacific

CSE 302: Object Oriented Programming II Lab: Visual and Web Design

Project Documentation

A Smart Web-Based Classroom Booking System for Students & Faculty

Submitted to

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Submitted by

Group Name: RoomEase

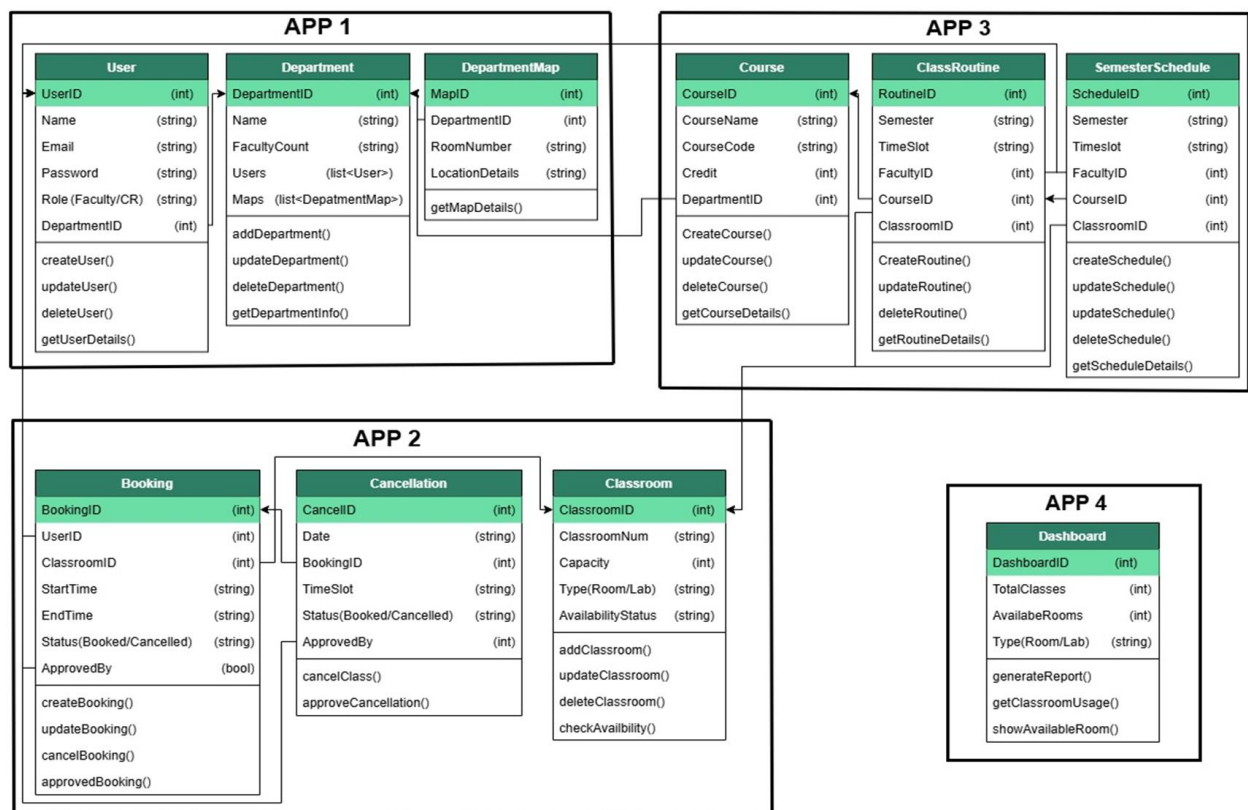
Group Number: 05

Name	Registration ID
Rifah Tasfia	22201126
Ankita Hossain Anushka	22201130
Md. Mahdi Hasan Nayeem	22201131
Rupasha Khan Tandra	22201136

Introduction

The Classroom Booking System is a web-based solution designed to optimize classroom utilization in universities. It addresses the problem of unused classrooms due to unexpected cancellations by providing real-time availability tracking, AI-based scheduling, and smart booking features. The system allows Class Representatives (CRs) and faculty to update schedules, while students can easily check room availability. The motivation behind this project is to enhance space utilization, reduce manual effort, and improve accessibility for students and faculty.

Class Diagram



Explanation of Classes:

- **User:** Handles authentication for CRs and faculty.
- **Classroom:** Stores details of classrooms (e.g., room number, capacity).
- **Schedule:** Manages semester-wise class schedules.
- **Booking:** Handles classroom bookings and cancellations with conflict detection.
- **ApprovalWorkflow:** Manages approval processes for cancellations.

Foreign Keys:

- ❖ Schedule references Classroom and User to link classes with rooms and updaters.
- ❖ Booking references Classroom and User to track bookings and users.

Project Structure

The system is built using Django and consists of the following apps:

- Users App: Handles authentication and user roles (CRs, faculty).
- ★ Models: User (extends Django's User model with role fields).
- Classrooms App: Manages classroom data.
- ★ Models: Classroom (fields: room_id, capacity, department).
- Schedules App: Manages class schedules.
- ★ Models: Schedule (fields: class_id, room_id, time_slot, status).
- Bookings App: Handles bookings and cancellations.
- ★ Models: Booking (fields: user_id, room_id, time_slot, status).

Installation and Setup

- ❖ Commands to setup the project in pycharm-

Open pycharm terminal & run the following commands sequentially-

- Git clone- <https://github.com/Nayeem131136/RoomEase.git>
- pip install django
- django-admin startproject RoomEase
- cd RoomEase
- python [manage.py](#) startapp accounts
- python [manage.py](#) startapp classrooms
- python [manage.py](#) migrate
- python [manage.py](#) createsuperuser
- After creating [models.py](#) in classrooms,
python [manage.py](#) startapp makemigrations
- python [manage.py](#) migrate
- python [manage.py](#) runserver

Then visit <http://127.0.0.1:8000/user/login> and <http://127.0.0.1:8000/admin>

- ❖ Outline the requirements.txt of the project:

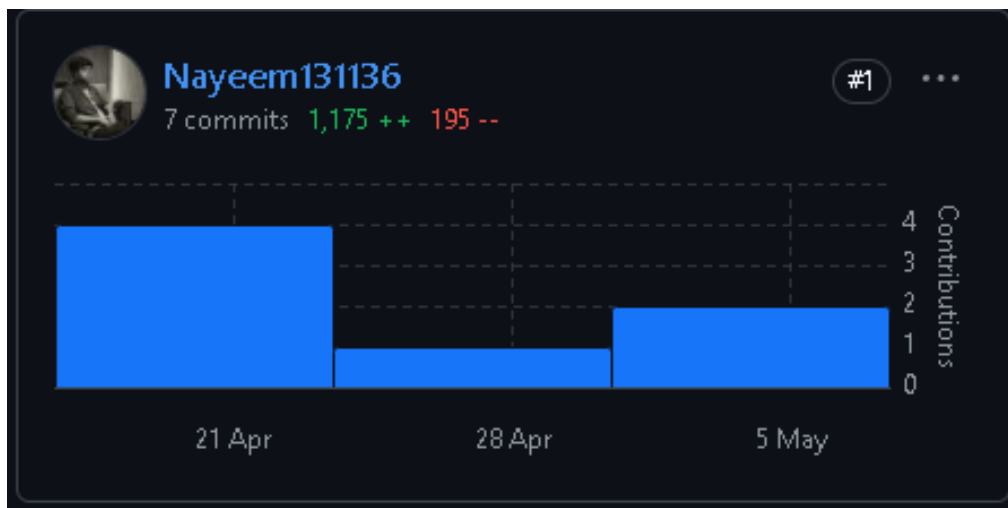
Package	Version	Latest version
asgiref	3.8.1	3.8.1
classroom	0.0.0.dev0	
django	5.2	5.2
pip	23.2.1	↑ 25.1.1
sqlparse	0.5.3	0.5.3
tzdata	2025.2	2025.2

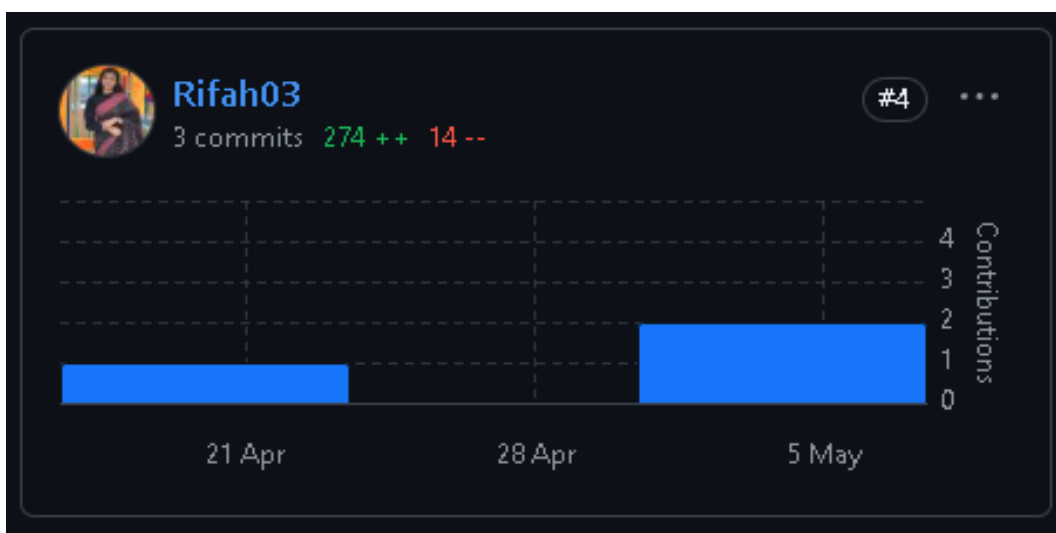
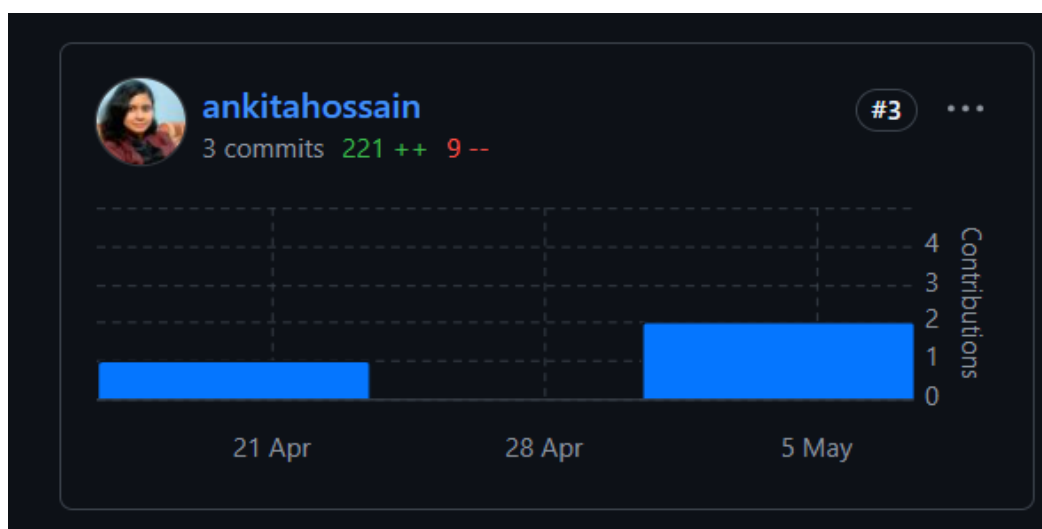
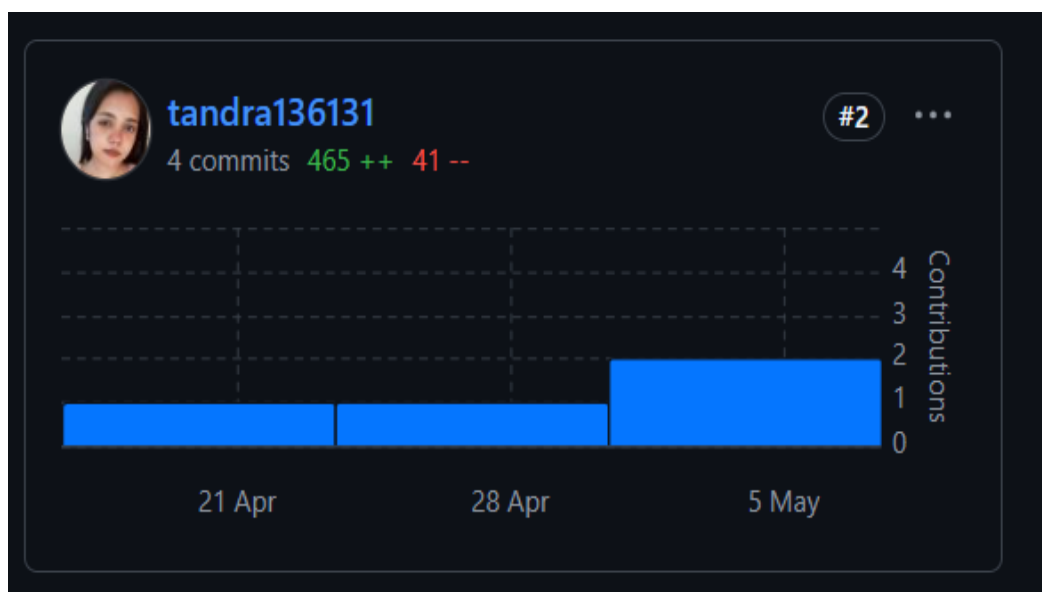
Contribution

- Github link:

<https://github.com/Nayeem131136/RoomEase>

- Contributions of four members:





➤ **contribution table:**

Table 1: Contribution of each member

SL	First Name	Reg ID	Github username	Number of Commit s	No of Lines added	No of Lines deleted
1.	Nayeem	22201131	Nayeem131136	7	1175	195
2.	Rupasha	22201136	tandra136131	4	465	41
3.	Rifah	22201126	Rifah03	3	274	14
4.	Anushka	22201130	ankitahossain	3	221	9

Screenshot

- **Login page**

The screenshot displays the login interface of the UAP RoomEase system. The browser's address bar shows the URL `127.0.0.1:8000/accounts/login/`. The page has a blue header with the text "UAP RoomEase" on the left and "Login Register" on the right. The main content area features a blue gradient box with the title "Login". Inside this box, there are two input fields: "Username" with the value "Nayeem131" and "Password" with masked characters. Below these fields are two buttons: "Login" and "Register". The footer of the page is blue and contains the text "DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING" and "UNIVERSITY OF ASIA PACIFIC". It also includes social media icons for Facebook and LinkedIn, and a "CONTACT" section with the address: "Department Administrative Office, Department of CSE, 74/A, Green Road, Dhaka - 1205, Bangladesh".

- Register page

UAP RoomEase

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF ASIA PACIFIC

Register

Username
Nayem131

Password

Re-Type Password

Register Login

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UNIVERSITY OF ASIA PACIFIC

f in

Papers & Publications UAP Webmail CONTACT
Department Administrative Offices

- DashBoard view

UAP RoomEase

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Logout

Welcome, uap

Manage classroom bookings efficiently

Room R702

Capacity: 50 students

View Full Schedule →

Today's Schedule (Sunday, May 4)

Time	Course	Section	Status	Action
8:00 AM - 9:20 AM	Available		Available	Book
9:30 AM - 10:50 AM	Available		Available	Book
11:00 AM - 12:20 PM	Available		Available	Book

🏠

Room R702

👤

Capacity: 50 students

➔

View Full Schedule

📅

Today's Schedule (Sunday, May 4)

Time	Course	Section	Status	Action
8:00 AM - 9:20 AM	Available		Available	📅 Book
9:30 AM - 10:50 AM	Available		Available	📅 Book
11:00 AM - 12:20 PM	Available		Available	📅 Book
12:30 PM - 1:50 AM	Available		Available	📅 Book
2:00 AM - 3:20 AM	BIE 301	A	Booked	🚫 Cancel
3:30 AM - 4:50 AM	Available		Available	📅 Book
5:00 AM - 6:20 AM	Available		Available	📅 Book

2:00 AM - 3:20 AM	BIE 301	A	Booked	🚫 Cancel
3:30 AM - 4:50 AM	Available		Available	📅 Book
5:00 AM - 6:20 AM	Available		Available	📅 Book

- Booking classroom

127.0.0.1:8000/rooms/R702/Sun/4/book/

UAP RoomEase

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Logout

Book Room R702

Booking Details:
Day: Sun
Time: 12:30 PM - 1:50 AM

Booking Reason:

need extra class for CSE 301

Please provide a detailed reason for your booking request (20-500 characters)

[Submit Request](#) [Cancel](#)

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[f](#) [in](#)

- Cancel classroom

127.0.0.1:8000/cancel/1/

UAP RoomEase

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Logout

Cancel Class

Class Details:
Room: R702
Day: Sun
Time: 2:00 AM - 3:20 AM
Course: BIE 301 A

Cancellation Reason:

Today teacher is not available

Please provide a valid reason for cancellation (20-500 characters)

[Confirm Cancellation](#) [Go Back](#)

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• Admin user

The screenshot shows the Django administration interface at the URL `127.0.0.1:8000/admin/auth/user/`. The page title is "Django administration" and the user is logged in as "UAP". The breadcrumb trail is "Home > Authentication and Authorization > Users".

On the left sidebar, under "AUTHENTICATION AND AUTHORIZATION", the "Users" link is highlighted. Under "CLASSROOMS", there are links for "Booking requests", "Daily schedules", "Rooms", and "Time slots", each with an "Add" button.

The main content area is titled "Select user to change". It features a search bar and a table of users. The table has columns for "USERNAME", "EMAIL ADDRESS", "FIRST NAME", "LAST NAME", and "STAFF STATUS". Two users are listed: "tandra" and "uap". The "uap" user is marked as a staff user with a green checkmark.

Below the table, it says "2 users". On the right, there is a "FILTER" sidebar with options to "Show counts", "By staff status" (All, Yes, No), "By superuser status" (All, Yes, No), and "By active" (All, Yes, No).

USERNAME	EMAIL ADDRESS	FIRST NAME	LAST NAME	STAFF STATUS
tandra	-	-	-	-
uap	-	-	-	✓

• Admin booking request_approval

The screenshot shows the Django administration interface at the URL `127.0.0.1:8000/admin/classrooms/bookingrequest/`. The page title is "Django administration" and the user is logged in as "UAP". The breadcrumb trail is "Home > Classrooms > Booking requests".

On the left sidebar, under "CLASSROOMS", the "Booking requests" link is highlighted. Under "AUTHENTICATION AND AUTHORIZATION", there are links for "Groups" and "Users", each with an "Add" button.

The main content area is titled "Select booking request to change". It features a search bar and a table of booking requests. The table has columns for "BOOKING REQUEST". Four booking requests are listed, all for "uap's request for R702 on Sun" at different time slots.

Below the table, it says "4 booking requests". On the right, there is an "ADD BOOKING REQUEST" button.

BOOKING REQUEST
uap's request for R702 on Sun at Slot 3 (11:00:00-12:20:00)
uap's request for R702 on Sun at Slot 2 (09:30:00-10:50:00)
uap's request for R702 on Sun at Slot 1 (08:00:00-09:20:00)
uap's request for R702 on Sun at Slot 1 (08:00:00-09:20:00)

127.0.0.1:8000/admin/classrooms/bookingrequest/7/change/

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

CLASSROOMS

Booking requests + Add

Daily schedules + Add

Rooms + Add

Time slots + Add

Change booking request

uap's request for R702 on Sun at Slot 3 (11:00:00-12:20:00) HISTORY

Requester: uap

Room: R702

Day: Sunday

Time slot: Slot 3 (11:00:00-12:20:00)

Reason: need extra class today

Status: Approved

SAVE Save and add another Save and continue editing Delete

127.0.0.1:8000/admin/classrooms/bookingrequest/

Django administration

WELCOME, UAP. VIEW SITE / CHANGE PASSWORD / LOG OUT

Home > Classrooms > Booking requests

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

CLASSROOMS

Booking requests + Add

Daily schedules + Add

Rooms + Add

Time slots + Add

The booking request "uap's request for R702 on Sun at Slot 3 (11:00:00-12:20:00)" was changed successfully.

Select booking request to change

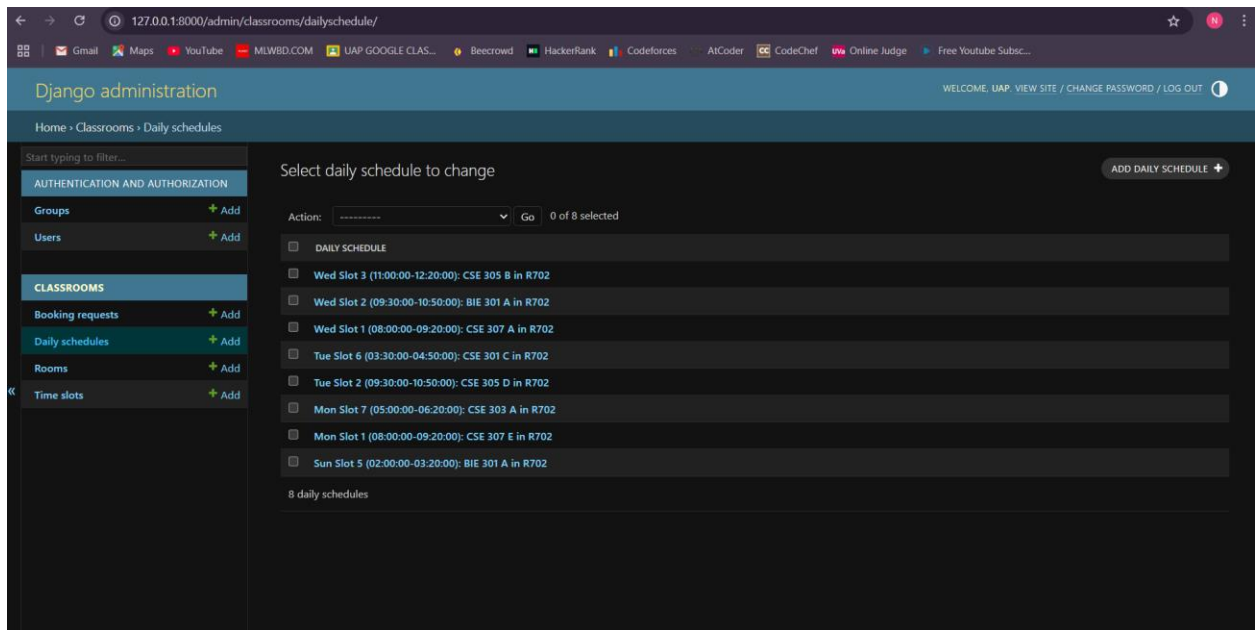
ADD BOOKING REQUEST +

Action: Go 0 of 4 selected

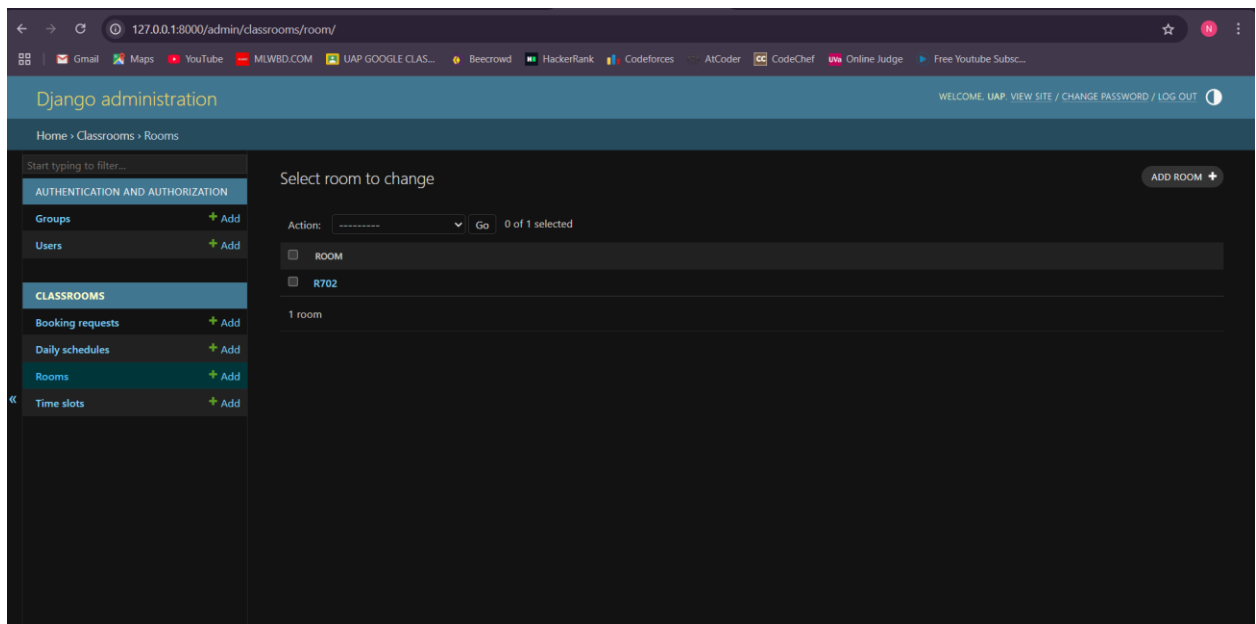
☐	BOOKING REQUEST
☐	uap's request for R702 on Sun at Slot 3 (11:00:00-12:20:00)
☐	uap's request for R702 on Sun at Slot 2 (09:30:00-10:50:00)
☐	uap's request for R702 on Sun at Slot 1 (08:00:00-09:20:00)
☐	uap's request for R702 on Sun at Slot 1 (08:00:00-09:20:00)

4 booking requests

• Admin daily schedule



• Admin rooms



- Admin time-slots

